A Novel Approach to GNN Explainability: Distilling Knowledge With Inter-Layer Alignment

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***Abstract*—Graphxx Neural Networks (GNNs) have made significant strides in the analysis and modeling of complex network data, particularly excelling in graph and node classification tasks. However, the "closed box" nature of GNNs impedes user understanding and trust, thereby restricting their broader application. This challenge has spurred a growing focus on demystifying GNNs to make their decision-making processes more transparent. Traditional methods for explaining GNNs often rely on selecting subgraphs and employing combinatorial optimization to generate understandable outputs. However, these methods are closely linked to the inherent complexity of GNNs, leading to higher explanation costs. To address this issue, we introduce a lower-complexity proxy model to explain GNNs. Our approach leverages knowledge distillation with inter-layer alignment, specifically targeting the challenge of over-smoothing and its detrimental impact on model explanation. Initially, we distill critical insights from complex GNN models into a more manageable proxy model. We then apply an inter-layer alignment-based distillation technique to ensure alignment between the proxy and the original model, facilitating the extraction of node or edge-level explanations within the proxy framework. We theoretically prove that the explanations derived from the proxy model are faithful to both the proxy and the original model. Additionally, we show that the upper bound of unfaithfulness between the proxy and the original model remains consistent when the distillation error is infinitesimal. This inter-layer alignment knowledge distillation technique enables the proxy model to retain the knowledge learning and topological representation capabilities of the original model to the greatest extent. Experimental evaluations on numerous real-world datasets confirm the effectiveness of our method, demonstrating robust performance.**

***Index Terms*—Graph neural network, explanation, knowledge distillation, inter-layer alignment.**

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I. Introduction

In recent years, Graph Neural Networks (GNNs) have made remarkable strides in the field of graph data processing, finding wide-ranging applications in social networks [1], bioinformatics [2], chemical molecules [3], recommendation systems [4] and other domains [5]. However, GNNs inherit the "closed box" characteristics of deep learning models [6], rendering their predicted results less trustworthy. Consequently, research on the interpretability of graph neural networks has gained significant attention.

GNNs leverage both node feature information and graph structure information by recursively propagating messages along the edges of the input graph. The expressiveness of GNNs is typically built on the highly nonlinear entanglement of irregular graph features [7], making it challenging to explain GNN models. Existing interpretability methods for GNNs can be broadly divided into two types: instance-level methods and model-level methods. Instance-level methods interpret the graph model by identifying crucial input features for prediction given an input graph, while model-level methods offer high-level insights and a general understanding of how deep graph models operate [8].

Among instance-level methods, GNNExplainer [9] stands out as a typical example, explaining predictions by learning soft masks of edge and node features and then optimizing these masks. At the instance level, methods can be grouped into decomposition and perturbation categories. GNN-LRP [38], an instance-level decomposition-based method, extends layer-wise relevance propagation to graph structures, tracing prediction relevance through multi-hop paths to form higher-order explanations beyond single nodes or edges.Similarly, GOAt [39] belongs to the decomposition family and analytically attributes model outputs to node and edge features without retraining auxiliary models, improving faithfulness and stability. In contrast, MixupExplainer [40] represents the perturbation-based line of work. It enhances the robustness of subgraph explanations against distribution shift through a generalized information bottleneck framework and mixup-based augmentation. Unlike perturbation-based explanation methods such as GNNExplainer, another popular approach is to identify important subgraph structures to explain GNNs, exemplified by SubgraphX [10]. On the other hand, a typical model-level method is XGNN [11], which explains GNNs at the model level by training a graph generator that maximizes a certain prediction. GMT [37] introduces an intrinsically interpretable GNN architecture based on subgraph multilinear extensions, where random subgraph sampling provides a closer approximation of the theoretical objective and yields better interpretability and generalization.

However, most current interpretability methods for GNNs are closely tied to their complexity, especially as the the number of layers increases. This complexity exacerbates the difficulty of interpreting GNN predictions because more layers mean more intermediate computational steps, making the model's inner workings more opaque. To address this issue and improve GNN interpretability, we introduce a novel method based on the inter-layer alignment knowledge distillation. Our aim is to transfer the knowledge from complex GNN models to simpler proxy models, thereby reducing computational costs while providing high-quality explanations. Specifically, our method transfers the knowledge of complex models into a lightweight proxy model, which is a simple GNN(e.g., S ^{2} GC [12]), through the concept of knowledge distillation [32]. The proxy model maintains consistency with the original model via an inter-layer alignment-based distillation approach, ensuring it simulates the behavior of the original model with high fidelity and accurately captures its key features. We validate our method on six common graph datasets, and the results demonstrate that our model surpasses most existing methods in terms of interpretability.

Actually, the "single-stage distillation" (which only allows students to imitate the final prediction results of the teacher) has obvious flaws: the knowledge of the teacher model is not only hidden in the "last layer of prediction", but also in the "feature representation of the middle layer" (for example, the middle layer of GNN learns the local structure and node correlation rules of the graph). And "layer by layer distillation" forces students to align each layer feature with the corresponding layer feature of the teacher (for example, if the teacher learns "strong correlation between nodes A and B" in the second layer, the student also needs to learn this rule in the second layer), which is equivalent to "breaking down the teacher's knowledge into multiple layers of details" and passing it on to students, avoiding the problem of students only learning" surface prediction results" and not learning "deep decision logic". This innovation prompts the core of this study lies more in addressing the "graph structure dependency" of GNN by leveraging inter-layer distillation. We have designed a layered strategy that adapts to its knowledge transfer: conventional layer by layer distillation focuses on feature alignment of "grid data (such as CNN images)", while the knowledge core of GNN is "topological information such as node associations and subgraph structures". Hence, directly applying conventional methods can easily lead to critical "topological knowledge loss", while our method primarily focus on the topological characteristics of GNN for knowledge transfer, it also considers other relevant factors. Based on the above discussions, the layer-layer alignment distillation designed in this paper can effectively address the "graph structure dependency" of GNN in the process of inter-layer knowledge transfer, thereby ensuring the credibility of the transferred results to a certain extent.

In summary, the main contributions of this paper are as follows:

1) We propose an improved S ^{2} GC as the proxy model for GNNs based on inter-layer loss alignment, controlling the loss difference of each layer between GNNs and their proxy models within a threshold. Using S ^{2} GC as the surrogate model for GNNs aligns more closely with the original GNN mechanisms compared to other proxy methods not based on GNNs.
2) We use the improved S ^{2} GC to compute the importance of each node and then generate importrant nodes for the original GNNs, effectively avoiding the over-smoothing problem and enhancing their explanations. Our experimental results achieve state-of-the-art performance.
3) This paper establishes a general framework for the explanation of complex models through simple proxy models, allowing the selection of proxy models based on the specific needs of different application fields.
4) We propose a general GNN explanation framework based on inter-layer alignment-based knowledge distillation, treating the GNN model as a closed box function. This method distills a simple agent model from the original GNN and further interprets it to obtain global explanation information, offering a novel and powerful approach to improving GNN interpretability.
5) We theoretically prove that the explanations derived from the proxy model are faithful to both the proxy and the original model, and we provide the upper bound of the unfaithfulness of the explanation to proxy and original model.

The remainder of the paper is structured as follows: Section II presents an overview of the related work and recent research achievements in relevant fields. In Section III, we introduce the background concepts and present our proposed framework for explainable GNNs based on inter-layer alignment-based knowledge distillation, including the formulation of related equations. Section IV delves into a detailed analysis of the theoretical foundations underpinning our model. Experimental results are presented in Section V, where we also discuss the implications of these findings. Finally, Section VI concludes our work and highlights potential directions for future research. Our code implementing the method of knowledge distilling with inter-layer alignment (short for DILA) is available online from https://github.com/ZXXSTUDENTGITHUB/DILA.

II. Preliminaries

We first review some basic knowledge that will be used in this paper.

A. Graph Neural Network

Deep learning's evolution has sparked increased interest in Graph Neural Networks (GNNs), such as Graph Convolutional Networks (GCN) [13], Graph Isomorphism Networks (GIN) [14], and Graph Attention Networks (GAT) [15]. GNNs operate within a message-passing framework that updates node features by aggregating neighboring information [16]. Given an undirected graph $\mathcal{G} = (\mathbf{V}, \mathbf{E})$, with adjacency matrix $\mathbf{A}$, degree matrix $\mathbf{D}$, and feature matrix $\mathbf{X}$, the message aggregation for the l-th layer of GNN is defined by

$$\mathbf{S}^l = \mathbf{X}^{l-1}\hat{\mathbf{X}}^l, \quad \mathbf{X}^l = M^l(\mathbf{S}^l),$$

where $X^t \in \mathbb{R}^{d_t \times n}$ represents the node feature matrix calculated from the t-th GNN layer and $X^0 = X$. $M^t(\cdot)$ represents the node feature change function at layer t.

We focus on three GNN models: GCN, SGC [17], and S ^{2} GC. GCN, widely used in various fields, operates by propagating node features through multiple layers. SGC simplifies GCN by collapsing activation functions into linear transformations, reducing parameters while improving accuracy. S ^{2} GC, a variant of GCN, employs a Markov diffusion kernel to aggregate information from larger neighborhoods, mitigating issues like over-smoothing. The prediction from an L-layer S ^{2} GC is:

$$\hat{Y} = \text{softmax}\left(\frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\widetilde{\mathbf{T}}^l\mathbf{X} + \alpha\mathbf{X}\right)\Theta\right).$$

Here, α balances self-information and neighborhood context, with $\widetilde{\mathbf{T}}^l$ being the normalized adjacency matrix.

B. GNNs Explanation

GNNs explanation mainly focus on tow key questions:

1) How can we explain a GNN model's results to users?
2) What specific knowledge, features, or subgraph node does the model leverage in its decision-making process?

Most of the current interpretable methods for GNNs are post-hoc explanation, whose aim is to uncover the rationality behind its outputs, decisions, or predictions by analyzing the already constructed model. In recent years, numerous approaches have been introduced to interpret GNNs, such as perturbation-based [9], decomposition-based [18] and model-level explanation mehods [11]. In the realm of GNN explanation, these strategies aim to highlight significant nodes, edges, features and important nodes that are pivotal in the model's reasoning. Among the various explanation methods, perturbation-based approaches are notably classical. As one of the notably classical post-hoc explanation methods [8], perturbation-based approaches ascertain the importance of input features by observing the variation in predictions under different input perturbations. For instance, GNNExplainer identifies crucial features by learning soft masks (values ranging between [0,1]) for edges and node features, facilitating explanation through mask optimization. This process involves creating a modified graph that emphasizes vital input information, followed by optimizing the mask to maximize the mutual information between the original graph and its modification.

Another type of significant post-hoc explanation method is gradient or feature-based approach, utilizing gradients or masked feature value maps to approximate the significance of inputs. Here, a larger gradient or feature value signifies the greater importance, with techniques like SA [19] utilizing the squared gradient values to rate the importance of various input features.

Decomposition-based methods, such as GCN-LRP [20], which allocate the model's prediction to different nodes as their importance scores, are also prevalent in interpreting GNNs.

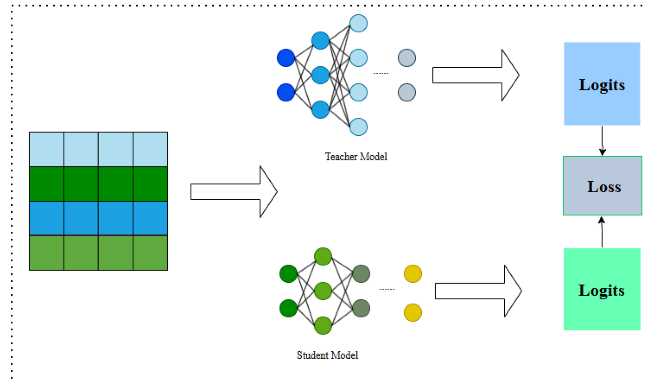


Fig. 1. The General Framework of Knowledge Distillation.

Meanwhile, proxy-based methods approximate complex models by employing simpler interpretable proxy models. PGM-Explainer, for instance, constructs a probabilistic graph model for instance-level explanation of GNNs.

Model-level interpretable methods strive to provide overarching insights and a higher-level understanding of GNNs. As a typical representative of model-level interpretable models, XGNN [11] trains a graph generator to produce graphs that maximize a target prediction, wherein graph generation is executed based on a reinforcement learning.

This refined narrative underscores the diverse methodologies adopted to interpret GNNs, highlighting their significance in enhancing the transparency and trustworthiness of GNN applications across various domains.

C. Knowledge Distillation

Although GNNs have achived the satisfactory performance in various graph learning tasks, they often require a large number of parameters in order to obtain better fitting ability, which increases the computational and interpretable costs.

As a learning paradigm, knowledge distillation (KD) enhances the resource efficient of more expressive but cumbersome teacher models [22]. The general framework of KD can be seen in Fig. 1, it learns the loss through aligning the teacher model with student model. KD is also a common compression solution of GNNs, it encourages the use of lightly weighted models (i.e., student models) to simulate the behavior of computationally expensive GNNs (i.e., teacher graph neural network models) [23]. In deep neural networks, each layer is responsible for learning different levels and degrees of feature representation. In recent years, the accuracy of the student model is almost equivalent to that of the teacher model in the field of GNNs, and it requires less computational cost. Xiang et al. [24] proposed a dedicated method for extracting knowledge from graph-less data by modeling the learning topology using a multivariate Bernoulli distribution for knowledge transfer. Chaitanya et al. [25] proposed graphical contrastive representation distillation based on contrastive learning, which implicitly preserves the global topology through aligning the student's node embeddings with the teacher's node embeddings in a shared representation space.

III. The Proposed Method

In this section, we propose a new post-hoc explanation method for GNNs based on inter-layer loss alignment, which consists of two main steps: knowledge distillation and explanation extraction. In the knowledge distillation stage, we use inter-layer alignment-based knowledge distillation method to obtain a linear GNN Ψ alternative from a complex GNN Φ. In the explanation extraction phase, we extract explanations directly from the simple alternative GNN Ψ instead of the complex GNN Φ. We argue that node that have a high influence on the output of GNN Ψ should be correlated with GNN Φ as long as GNN Ψ is a good approximation of the GNN Φ. Therefore, the explanation of the proxy model should also be able to explain the original GNN model itself well.

Our method introduces a novel inter-layer knowledge distillation framework for graph neural networks (GNNs), which focuses on transferring knowledge between different layers of the model. Different from traditional methods, our framework improves the distillation process by aligning the losses between each layer of the GNN and its surrogate model S ^{2} GC. Compared with other methods, this inter-layer alignment ensures that the surrogate model learns the hierarchical knowledge inherent in the original GNN, thereby more effectively preserving its structural characteristics. By proposing a common framework, we avoid the over-smoothing problem often encountered in GNNs, where important node features are lost during the aggregation process. Our experimental results show that the framework has state-of-the-art performance in GNN interpretation, providing a powerful tool for enhancing model transparency and understanding complex graph-based relationships.

A. Knowledge Distillation With Inter-Layer Alignment

The knowledge distillation aims to adjust the parameters of the proxy model α so that its predictions match those of the original model β. We adopt an inter-level alignment approach for knowledge distillation with the aim of preserving the proxy model's ability to learn and represent knowledge from the original model as faithfully as possible.

We first need to obtain a simple proxy model α for complex GNN by using inter-layer alignment-based knowledge distillation method, and here we choose S ^{2} GC as the simple alternative α to do the explanation. The following will give the learning process of the proxy model from a new insight.

Given a graph $\mathcal{G} = (\mathbf{A}, \mathbf{X})$, the prediction of the proxy model for $\mathcal{G}$ can be obtained according to $\hat{Y}$. Through extending the Markov diffusion kernel approach, S ^{2} GC designs a filter to capture the global and context of each node by combining the trade-offs of low-pass and high-pass filter bands, it exploits the cascade of progressively increasing contexts while limiting transition smoothing by giving proportionally greater weight to each node's nearest neighbors.

Deep neural networks excel at learning multiple levels of representation, either the output of the last layer or the output of the middle layer, i.e., the feature map, can be used as a source of knowledge for supervised student model training. Specifically, inter-layer alignment-based knowledge from the middle layer is a good complement to distillation methods based on the output of the last layer. The main idea of inter-layer alignment-based knowledge distillation is to directly match the feature outputs from the intermediate layers of teachers and students.

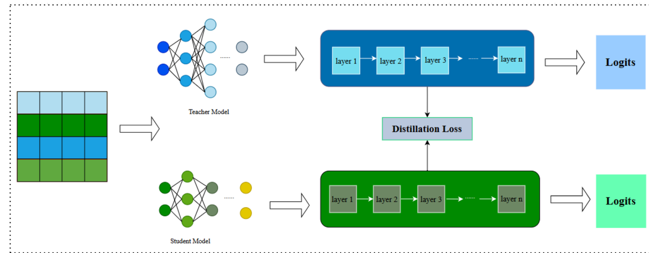


Fig. 2. Inter-Layer Alignment-based Knowledge Distillation.

The output of the last layer usually contains task-relevant information at the highest level, while the feature maps of the middle layer capture the more underlying, concrete features of the data. Thus, it can improve the learning performance of the model on complex tasks through utilizing both the outputs of the last and middle layers as knowledge sources. In feature-level knowledge distillation, we directly compare the feature outputs generated by the original and proxy models at the middle layer. Such a matching process can ensure that the proxy model not only learns a high-level abstract representation relevant to the task, but also captures underlying features that are similar to those of the original model. This fine-grained feature matching is contribute to conveying more detailed information about the data, improving the model's task understanding.

It is worth noting that it can provide more comprehensive and in-depth knowledge guidance to the proxy model according to the combination of the last layer of output and the middle layer of features in together, based on which it can optimize the training process of the model to make it perform better in various complex tasks. Furthermore, this combination preserves the learning ability of the proxy model from the perspective of topology. This combined use of different levels of representation provides more comprehensive and multi-level learning guidance for the model, and it preserves the learning ability of the proxy model to the original model from the perspective of topological consistency.

Based on above discussions, the knowledge distillation loss of the i-th layer L $L_{\text{KL}}^{(i)}$ is given as

$$L_{\text{KL}}^{(i)} = \sum_{x \in \mathbf{X}} \text{softmax}(s_i(x)) \log \left(\frac{\text{softmax}(s_i(x))}{\text{softmax}(t_{f(j)}(x))} \right), \quad (1)$$

where $\text{softmax}(s_i(x))$ denotes the probability distribution of the i-th layer's output of the proxy model processed by the softmax function, $f(\cdot)$ is a mapping function that projects the j-th layers of the original model to the same dimension as those of the i-th layer of proxy model, $\log(\cdot)$ calculates the difference of the probability distributions between the output of the corresponding layers of the original model and proxy model. The model for knowledge distillation with inter-layer alignment is shown in Fig. 2.

The overall knowledge distillation loss with inter-layer alignment that measured by *KL* divergence can be expressed as

$$L_{\mathrm{KL}} = \frac{1}{L_S}\sum_{i=1}^{L_S} L_{\mathrm{KL}}^{(i)}, \tag{2}$$

where L_S refers to the number of latent layers in the proxy model. The knowledge distillation losses for all layers are averaged to obtain the overall distillation loss. L_{KL} evaluates the learning performance of the proxy model to original model from a global perspective.

B. Explanation Extraction With Proxy Model

We identify a subgraph of the input graph $\mathcal{G}$ to obtain an explanation for a given prediction, wherein the subgraph consists of those nodes that have the greatest influence on prediction. The explanation $\mathcal{E}$ is denoted as an n-dimensional vector of importance scores, one corresponding to each node in the top node set $\mathbf{V}$. Thus, the explanation vector $\mathcal{E}$ can be represented as

$$\mathcal{E} = [\mathcal{E}_1, \mathcal{E}_2, \ldots, \mathcal{E}_p],$$

where $\mathcal{E}_i (i = 1, \ldots, p)$ refers to interpreting the predictions of machine learning models, which can help to build transparent and trustworthy AI systems. It can be seen as an optimization task to find models' explanations. Specifically, we choose to represent explanations as 0-1 vectors of binary weights. Its target is to minimize the number of logarithms associated with the predictions, rather than the L_2 paradigm of the graph that selectively masks node features according to the explanation vector $\mathcal{E}$. The explanation vector $\mathcal{E}$ is a binary vector whose elements denotes the importance of nodes to the predictions.

The optimization process involves adjusting the binary weights of the explanation vectors in order to strike a balance between maintaining model accuracy and simplifying explanation. Assigning a weight of 1 to nodes considered important and a weight of 0 to nodes considered irrelevant, the explanation vector guides the explanation of the model's decision boundaries.

In summary, the process of finding explanations involves formulating the problem as an optimization task, where explanation vectors are optimized to minimize the difference between model predictions with and only with explanations,

$$\min_{\mathcal{E}\in\{0,1\}^n} \left\| \left(\frac{1}{L}\sum_{l=1}^{L} \left((1-\alpha)\widetilde{\mathbf{T}}^l \operatorname{diag}(\mathcal{E})\mathbf{X} + \alpha\mathbf{X} \right) \Theta \right.\right. \\ \left.\left. - \frac{1}{L}\sum_{l=1}^{L} \left((1-\alpha)\widetilde{\mathbf{T}}^l\mathbf{X} + \alpha\mathbf{X} \right) \Theta \right) \right\|_2, \tag{3}$$

where $\operatorname{diag}(\mathcal{E})$ is a diagonal matrix with elements of 1 or 0 on the diagonal, $\|\cdot\|_2$ denotes the L_2 norm.

C. Complexity Analysis

In this subsection, we discuss the complexity analysis of our proposed method in terms of model distillation and explanation extraction.

1) Model Distillation Process: The computational time of the model distillation is mainly influenced by three factors: the number of latent layers, the number of nodes in each layer, and the total number of parameters in the model. Specifically, it needs to carry out forward propagation calculations for both the original and proxy models to evaluate the distillation loss in the distillation process. The computational complexity of this part is directly related to the number of layers and the number of nodes per layer: an increase in either the number of layers or the number of nodes per layer will linearly increase the computational load, hence forward propagation calculations of the time complexity can be represented as $O(L \cdot N)$. Subsequently, the time complexity of the backward propagation and parameter update phase is mainly related to the number of parameters in the model, because the gradients are calculated based on the loss function, and the performance of the proxy model is optimized by the model's parameters. An increase in the number of parameters will linearly increase the computational load, therefore the time complexity of this part is $O(P)$. Thus, the overall time complexity of the model distillation process is $O(L \cdot N) + O(P)$, which indicates when conducting the model distillation, one must consider the number of layers and the number of nodes per layer, as well as the number of model parameters.

2) Explanation Extraction Process: The time complexity of the explanation extraction directly determines the efficiency of the model. Explanation extraction consists of two main parts: one is the explanation extraction and loss calculation performed in each iteration, and the other is the gradient back propagation and parameter updating. Firstly, the explanation extraction and loss calculation are executed in every iteration, where the former refers to the creation of explanations for the model's decisions based on the current model parameters, while the latter is used to evaluate the quality or accuracy of these explanations. The time complexity of these two steps is directly proportional to the number of iterations, as they need to be performed once in each iteration. Therefore, the time complexity of the explanation extraction is at least $O(num\_epochs)$, meaning that the total computation time increases linearly with the number of iterations. Next, we need to consider the steps of gradient back propagation and parameter updates. The time complexity of the gradient back propagation and parameter updating does not depend on the number of iterations but is directly related to the number of parameters in the model. Specifically, gradient back propagation requires calculating the gradient for each parameter, and parameter updating necessitate adjusting the value of each parameter based on these gradients. Therefore, the time complexity of this part can be considered as $O(p)$. Thus, the time complexity of the explanation extraction is determined by two factors: the number of iterations and the number of parameters in the model. Therefore, the overall time complexity of the explanation generation process can be approximated as $O(num\_epochs \cdot P)$, which means the total computation time increases with the increase in the number of iterations and the number of parameters.

IV. Theoretical Analysis

In this section, we investigate the error bounds of our proposed method with theoretical quantitative analysis: a good explanation from the proxy model Ψ is also a reliable and faithful explanation of the original model Φ.

Let $\mathcal{G}_u$ be a subgraph of $\mathcal{G}$, which consists of the L-hop neighborhood around node u, $\mathcal{E}_u$ an explanation for u. We say $\mathcal{E}_u$ is faithful [26], [31] in terms of Φ if it satisfies: i) the outputs of Φ yield nearly identical predictions for u, regardless of whether the nodes of $\mathcal{G}_u$ are weighted using $\mathcal{E}_u$ or not; and ii) the same remains valid when perturbing $\mathcal{G}_u$ slightly. $\mathcal{G}_u$ can be perturbed by adding noise to the u's features or randomly rewiring the incident edges of node u' [31]. In the current work, we consider perturbations to node features.

Definition 1: (Faithfulness) [31]. Let $\mathcal{K}$ be a set of perturbations of $\mathcal{G}_u$, $\mathcal{E}_u$ an explanation of $\mathcal{G}_u$ with respect to model f, if

$$\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u\in\mathcal{K}\cup\{\mathcal{G}_u\}}\|f(\mathcal{G}'_u)-f(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\leq\delta, \tag{4}$$

then we call $\mathcal{E}_u$ is faithful to f. $\mathcal{G}'_u$ is the graph obtained after perturbing $\mathcal{G}_u$ based on node features, t is a function that applies the explanation $\mathcal{E}_u$ to the graph $\mathcal{G}'_u$, and δ is a small constant.

Lemma 1: Let $\mathcal{K}$ be a set of perturbations and u a node, the unfaithfulness bound of the explanation $\mathcal{E}_u$ from Ψ w.r.t. to the prediction $Y_u(\Psi_\Theta)$ of node u is expressed as

$$\begin{aligned}&\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u\in\mathcal{K}\cup\{\mathcal{G}_u\}}\|\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\&\leq\gamma\max_{l\in\{1,\ldots,T\}^n}\left\|\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}^l_u\right\|_2,\end{aligned} \tag{5}$$

where $\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}^l_u$ is the u-th row of $\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}^l (l\in\{1,2,\ldots,L\})$ corresponds to layer l, and $\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}^l=\widetilde{\mathbf{T}}^l-\widetilde{\mathbf{E}}^l$ is the adjacency matrix of those unimportant nodes, that is, it is difference matrix between before and after the explanation $\mathcal{E}_u$ is applied; γ is a constant associated with the model weights Θ, node features $\mathbf{X}$, and perturbation ϵ.

Proof: As softmax is Lipschitz continuous with respect to the $\|\cdot\|_2$ norm [29], according to Definition 1 and taking into account the induced matrix norm compatibility property [30], setting $\widetilde{\mathbf{T}}^l\operatorname{diag}(\mathcal{E})=\widetilde{\mathbf{E}}^l$ for node u, we can get

$$\begin{aligned}&\|\Psi(\mathcal{G}_u)-\Psi(t(\mathcal{G}_u,\mathcal{E}_u))\|_2\\
&=\Bigg\|\sigma\left(\frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\widetilde{\mathbf{T}}^l_u\mathbf{X}+\alpha\mathbf{X}\right)\Theta\right)\\
&\quad-\sigma\left(\frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\widetilde{\mathbf{T}}^l_u\operatorname{diag}(\mathcal{E})\mathbf{X}+\alpha\mathbf{X}\right)\Theta\right)\Bigg\|_2\\
&\leq\Bigg\|\frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\widetilde{\mathbf{T}}^l_u\mathbf{X}+\alpha\mathbf{X}\right)\Theta\\
&\quad-\frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\widetilde{\mathbf{E}}^l_u\mathbf{X}+\alpha\mathbf{X}\right)\Theta\Bigg\|_2\\
&=\left\|\frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\left(\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right)\mathbf{X}\right)_u\Theta\right\|_2\\
&\leq(1-\alpha)\frac{1}{L}\sum_{l=1}^{L}\left\|\left(\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right)\mathbf{X}\Theta\right\|_2\\
&\leq(1-\alpha)\frac{1}{L}\sum_{l=1}^{L}\left\|\left(\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right)\right\|_2\|\mathbf{X}\Theta^T\|_2\\
&=(1-\alpha)\left\|(\mathbf{X}\Theta)^T\right\|_2\frac{1}{L}\sum_{l=1}^{L}\left\|\left(\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right)\right\|_2\\
&\leq(1-\alpha)\left\|(\mathbf{X}\Theta)^T\right\|_2\max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right\|_2.\end{aligned}$$

Similarly, we get the unfaithfulness of the explanation $\mathcal{E}_u$ with respect to Ψ by adding a perturbation Σ'_u to $\mathbf{X}$, where $\mathbf{\Sigma}'_u\in\mathbf{R}^{n\times d}$ with the same dimensions with $\mathbf{X}\in\mathbf{R}^{n\times d}$,

$$\begin{aligned}&\|\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\&\leq(1-\alpha)\left\|\left[(\mathbf{X}+\mathbf{\Sigma}'_u)\,\Theta\right]^T\right\|_2\max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right\|_2.\end{aligned}$$

It follows the mean over $\mathcal{K}\cup\{\mathcal{G}_u\}$ that

$$\begin{aligned}&\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u\in\mathcal{K}\cup\{\mathcal{G}_u\}}\|\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\
&\leq\frac{1}{|\mathcal{K}|+1}\Bigg[(1-\alpha)\,\|(\mathbf{X}\Theta)^\mathsf{T}\|_2\max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right\|_2\\
&\quad+\sum_{\mathcal{G}'_u\in\mathcal{K}}\|[(\mathbf{X}+\Sigma'_u)\Theta]^\mathsf{T}\|_2\max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right\|_2\Bigg]\\
&\leq(1-\alpha)\,\|\Theta^T\|_2\max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right\|_2\cdot\\
&\quad\frac{1}{|\mathcal{K}|+1}\left(\|\mathbf{X}^T\|_2+\sum_{\mathcal{G}'_u\in\mathcal{K}}\|(\mathbf{X}+\mathbf{\Sigma}'_u)^\mathsf{T}\|_2\right)\\
&\leq(1-\alpha)\,\|\Theta^T\|_2\max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u\right\|_2\cdot\\
&\quad\left(\|\mathbf{X}^T\|_2+\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u\in\mathcal{K}}\|\mathbf{\Sigma}'_u\|_2\right)\\
&=(1-\alpha)\gamma_{\Theta,X,\Sigma'_u}\max_{l\in\{1,\ldots,T\}^n}\left\|\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}^l_u\right\|_2,(\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}^l_u=\widetilde{\mathbf{T}}^l_u-\widetilde{\mathbf{E}}^l_u).\end{aligned}$$

When $\mathbf{\Sigma}'_u$ is limited, the constant γ may not depend on $\mathbf{\Sigma}'_u$.

Lemma 1 offers an account and an upper bound on the unfaithfulness of explanation $\mathcal{E}_u$ provided by proxy model Ψ.

When apply the explanation $\mathcal{E}_u$ obtained by proxy model Ψ to the original model Φ, the upper bound of unfaithfulness is given by the following theorem.

Theorem 1: Let $\mathcal{K}$ be a set of perturbations and u a node, and original model Φ is affected by a distillation loss of β, then the unfaithfulness bound of the explanation $\mathcal{E}_u$ to the Φ's prediction with respect to u is given as follows:

$$\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Phi(\mathcal{G}'_u)-\Phi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2 \leq \gamma \max_{l\in\{1,\ldots,T\}^n} \left\|\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l\right\|_2 + 2\beta. \tag{6}$$

Proof: We know

$$\begin{aligned}&\|\Phi(\mathcal{G}'_u)-\Phi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\ &=\|\Phi(\mathcal{G}'_u)-\Psi(\mathcal{G}'_u)+\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\\ &\quad+\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))-\Phi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2.\end{aligned}$$

According to Lemma 1 and triangle inequality, it follows the results that

$$\begin{aligned}&\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Phi(\mathcal{G}'_u)-\Phi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\ &\leq \frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Phi(\mathcal{G}'_u)-\Psi(\mathcal{G}'_u)\|_2\\ &\quad+\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\ &\quad+\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))-\Phi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\ &\leq (1-\alpha)\gamma_{\Theta,X,\Sigma'_u} \max_{l\in\{1,\ldots,T\}^n} \left\|\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l\right\|_2 + 2\beta.\end{aligned}$$

Theorem 1 is valid based on the distillation error α, it follows an upper bound same to Lemma 1 when Ψ approaches Φ infinitely. It means that the unfaithfulness bound of explainer to original model Φ is controlled by the degree of approximation of Ψ to Φ.

The following lemma gives the probabilistic nature of the noise about the bounds of Lemma 1.

Lemma 2: Let $\mathcal{K}$ be a set of perturbations and u a node. Suppose that the perturbations $\varepsilon_{u'}$ follows a normal distribution, i.e., $\varepsilon_{u'} \sim \mathcal{N}(0,\sigma^2)$, then the following unfaithfulness bound of the explanation $\mathcal{E}_u$ w.r.t. the prediction $Y_u(\Psi_\Theta)$ of node u holds with probability at least $1-p$ over $\mathcal{K}\cup\{\mathcal{G}_u\}$,

$$\begin{aligned}&P\left(\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2 \leq \epsilon\right)\\ &\geq F_{\chi^2(|\mathcal{K}|nd)}\left(\frac{\epsilon-(1-\alpha)\gamma_{\Theta,X}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l}{\frac{(1-\alpha)\sigma\gamma_\Theta}{|\mathcal{K}|+1}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l}-|\mathcal{K}|\right).\end{aligned}$$

where $p = 1 - F_{\chi^2(|\mathcal{K}|nd)}\left(\frac{\epsilon-(1-\alpha)\gamma_{\Theta,X}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l}{\frac{(1-\alpha)\sigma\gamma_\Theta}{|\mathcal{K}|+1}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l}-|\mathcal{K}|\right)$, and $\gamma_{\Theta,X}$ and γ_Θ are constants, n and d are the dimensions of $\mathbf{X}$, that is, $\mathbf{X}\in\mathbf{R}^{n\times d}$.

Proof: According to the proof of Lemma 1, we get

$$\begin{aligned}&\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\ &\leq (1-\alpha)\|\Theta^T\|_2 \max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}_u^l-\widetilde{\mathbf{E}}_u^l\right\|_2\cdot\\ &\quad\left(\|\mathbf{X}^T\|_2+\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u\in\mathcal{K}}\|\mathbf{\Sigma}'_u\|_2\right).\end{aligned} \tag{7}$$

The following result is obtained from [26], that is,

$$\sum_{\mathcal{G}'_u\in\mathcal{K}}\|\mathbf{\Sigma}'_u)^\mathsf{T}\|_2 \leq \sigma|\mathcal{K}|+\sigma Q,$$

with $\epsilon_{j,i} \sim \mathcal{N}(0,\sigma^2)$ and $Q \sim \chi^2(|\mathcal{K}|nd)$ for each element $\epsilon_{j,i}$ in Σ'_u and $Q=\sum_{i=1}^{nd} Z^2_{u',i}$, respectively, wherein $Z_{u',i} \sim \mathcal{N}(0,1)$.

Thus, if $\chi^2_p(|\mathcal{K}|nd)$ is the p-percentile of Q, then the probability of the last inequality of (7) is p :

$$\begin{aligned}&\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Psi(\mathcal{G}'_u)-\Psi(t(\mathcal{G}'_u,\mathcal{E}_u))\|_2\\ &\leq (1-\alpha)\|\Theta^T\|_2 \max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}_u^l-\widetilde{\mathbf{E}}_u^l\right\|_2\cdot\\ &\quad\left(\|\mathbf{X}^T\|_2+\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u\in\mathcal{K}}\|\mathbf{\Sigma}'_u\|_2\right)\\ &\leq (1-\alpha)\|\Theta^T\|_2 \max_{l\in\{1,\ldots,T\}^n}\left\|\widetilde{\mathbf{T}}_u^l-\widetilde{\mathbf{E}}_u^l\right\|_2\cdot\\ &\quad\left(\|\mathbf{X}^T\|_2+\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u\in\mathcal{K}}\sigma(|\mathcal{K}|+Q)\right)\\ &\leq (1-\alpha)\gamma_{\Theta,X}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l+\\ &\quad\frac{(1-\alpha)\gamma_\Theta}{|\mathcal{K}|+1}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l\sigma\left(|\mathcal{K}|+\chi^2_p(|\mathcal{K}|nd)\right).\end{aligned} \tag{8}$$

When σ approaches to zero, the error caused by perturbation is negligible. Setting the right side of (8) to ϵ, then we get

$$\chi^2_p(|\mathcal{K}|nd)=\frac{\epsilon-(1-\alpha)\gamma_{\Theta,X}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l}{\frac{(1-\alpha)\sigma\gamma_\Theta}{|\mathcal{K}|+1}\max_{l\in\{1,\ldots,T\}^n}\underset{\mathcal{E}_u}{\Delta}\widetilde{\mathbf{A}}_u^l}-|\mathcal{K}|.$$

According to the definition of Chi-square cumulative distribution function $F_{\chi^2(|\mathcal{K}|nd)}$ of Q and

$$P(X\leq\chi^2_p(|\mathcal{K}|nd))=\mathcal{F}_{\chi^2_{|\mathcal{K}|nd}}\left(\chi^2_p(|\mathcal{K}|nd)\right)=1-p,$$

it follows $p = 1 - \mathcal{F}_{\chi^2_{|\mathcal{K}|nd}}(\chi^2_p(|\mathcal{K}|nd))$. Therefore, we have

$$P\left(\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Psi(\mathcal{G}'_u) - \Psi(t(\mathcal{G}'_u, \mathcal{E}_u))\|_2 \le \epsilon\right)$$
$$\ge F_{\chi^2(|\mathcal{K}|nd)}\left(\frac{\epsilon - (1-\alpha)\,\gamma_{\Theta,X}\max\limits_{l\in\{1,\ldots,T\}^n}\Delta\widetilde{\mathbf{A}}^l_u{}_{\mathcal{E}_u}}{\frac{(1-\alpha)\sigma\gamma_\Theta}{|\mathcal{K}|+1}\max\limits_{l\in\{1,\ldots,T\}^n}\Delta\widetilde{\mathbf{A}}^l_u{}_{\mathcal{E}_u}} - |\mathcal{K}|\right).$$

The following theorem describes the probabilistic nature of the noise about the bounds of Theorem 1.

Theorem 2: Let $\mathcal{K}$ be a set of perturbations and u a node, the original model Φ is affected by a distillation loss of β. Assume that the perturbation $\varepsilon_{u'}$ follows the normal distribution, i.e., $\varepsilon_{u'} \sim \mathcal{N}(0, \sigma^2)$, then the following unfaithfulness bound of the explanation $\mathcal{E}_u$ to the Φ's prediction about u holds with probability $1-p$ over $\mathcal{K}\cup\{\mathcal{G}_u\}$,

$$P\left(\frac{1}{|\mathcal{K}|+1}\sum_{\mathcal{G}'_u \in \mathcal{K}\cup\{\mathcal{G}_u\}} \|\Phi(\mathcal{G}'_u) - \Phi(t(\mathcal{G}'_u, \mathcal{E}_u))\|_2 \le \epsilon\right)$$
$$\ge F_{\chi^2(|\mathcal{K}|nd)}\left(\frac{\epsilon - 2\beta\,(1-\alpha)\,\gamma_{\Theta,X}\max\limits_{l\in\{1,\ldots,T\}^n}\Delta\widetilde{\mathbf{A}}^l_u{}_{\mathcal{E}_u}}{\frac{(1-\alpha)\sigma\gamma_\Theta}{|\mathcal{K}|+1}\max\limits_{l\in\{1,\ldots,T\}^n}\Delta\widetilde{\mathbf{A}}^l_u{}_{\mathcal{E}_u}} - |\mathcal{K}|\right).$$

where

$$p = 1 - F_{\chi^2(|\mathcal{K}|nd)}$$
$$\left(\frac{\epsilon - 2\beta - (1-\alpha)\,\gamma_{\Theta,X}\max\limits_{l\in\{1,\ldots,T\}^n}\Delta\widetilde{\mathbf{A}}^l_u{}_{\mathcal{E}_u}}{\frac{(1-\alpha)\sigma\gamma_\Theta}{|\mathcal{K}|+1}\max\limits_{l\in\{1,\ldots,T\}^n}\Delta\widetilde{\mathbf{A}}^l_u{}_{\mathcal{E}_u}} - |\mathcal{K}|\right),$$

$\gamma_{\Theta,X}$ is a constant that depends on Θ and $\mathbf{X}$, and γ_Θ is a constant refers to Θ, n and d are the dimensions of $\mathbf{X}$, that is, $\mathbf{X} \in \mathbf{R}^{n\times d}$.

The proof of Theorem 2 can be obtained from Lemma 2.

It is worth mentioning that the loss landscape of our extraction problem is exclusively determined by the configuration or structure of Ψ, disregarding the original GNN model Φ used in GNNExplainer. As Ψ is an S ^{2} GC, Theorem 3 establishes that (3) is convex function. Consequently, it can achieve global optima by employing gradient-based algorithms, such as those commonly used in optimization.

Theorem 3: The objective function of (3) is convex.

Proof: According to (3), we have

$$f(\mathcal{E}) = \left\|\left(\frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\widetilde{\mathbf{T}}^l_i\,\mathrm{diag}(\mathcal{E})\mathbf{X} + \alpha\mathbf{X}\right)\Theta - \frac{1}{L}\sum_{l=1}^{L}\left((1-\alpha)\widetilde{\mathbf{T}}^l_i\mathbf{X} + \alpha\mathbf{X}\right)\Theta\right)\right\|_2^2$$
$$= \frac{(1-\alpha)^2}{L^2}\cdot\left\|\sum_{l=1}^{L}\left(\widetilde{\mathbf{T}}^l_i\,\mathrm{diag}(\mathcal{E})\mathbf{X} - \widetilde{\mathbf{T}}^l_i\mathbf{X}\right)\Theta\right\|_2^2$$
$$= \frac{(1-\alpha)^2}{L^2}\mathrm{Tr}\left(\left(\sum_{l=1}^{L}\left(\widetilde{\mathbf{T}}^l_i\,\mathrm{diag}(\mathcal{E})\mathbf{X}\Theta - \widetilde{\mathbf{T}}^l_i\mathbf{X}\Theta\right)\right)\cdot\left(\sum_{l=1}^{L}\left(\widetilde{\mathbf{T}}^q_i\,\mathrm{diag}(\mathcal{E})\mathbf{X}\Theta - \widetilde{\mathbf{T}}^q_i\mathbf{X}\Theta\right)\right)^T\right)$$
$$= \mathrm{Tr}\left(\sum_{l,q=1}^{L}\widetilde{\mathbf{T}}^l_i\,\mathrm{diag}(\mathcal{E})\mathbf{X}\Theta\Theta^T\mathbf{X}^T(\mathrm{diag}(\mathcal{E}))^T(\widetilde{\mathbf{T}}^p_i)^T\right)$$
$$- 2\mathrm{Tr}\left(\sum_{l,p=1}^{L}\widetilde{\mathbf{T}}^l_i\,\mathrm{diag}(\mathcal{E})\mathbf{X}\Theta\Theta^T\mathbf{X}^T(\widetilde{\mathbf{T}}^p_i)^T\right)$$
$$+ \mathrm{Tr}\left(\sum_{l,p=1}^{L}\widetilde{\mathbf{T}}^l_i\mathbf{X}\Theta\Theta^T\mathbf{X}^T(\widetilde{\mathbf{T}}^p)^T\right)$$
$$= \sum_{l,p=1}^{L}\mathrm{Tr}\left(\widetilde{\mathbf{T}}^l_i\,\mathrm{diag}(\mathcal{E})\mathbf{X}\Theta\Theta^T\mathbf{X}^T(\mathrm{diag}(\mathcal{E}))^T(\widetilde{\mathbf{T}}^p_i)^T\right)$$
$$- 2\sum_{l,p=1}^{L}\mathrm{Tr}\left(\widetilde{\mathbf{T}}^l_i\,\mathrm{diag}(\mathcal{E})\mathbf{X}\Theta\Theta^T\mathbf{X}^T(\widetilde{\mathbf{T}}^p_i)^T\right)$$
$$+ \sum_{l,p=1}^{L}\mathrm{Tr}\left(\widetilde{\mathbf{T}}^l_i\mathbf{X}\Theta\Theta^T\mathbf{X}^T(\widetilde{\mathbf{T}}^p)^T\right).$$

The first partial derivative of $f(\mathcal{E})$ with respect to $\mathrm{diag}(\mathcal{E})$ is

$$\frac{\partial f(\mathcal{E})}{\partial \mathrm{diag}(\mathcal{E})} = \sum_{l,p=1}^{L}(\widetilde{\mathbf{T}}^l_i)^T\widetilde{\mathbf{T}}^p_i\mathrm{diag}(\mathcal{E})\mathbf{X}\Theta\Theta^T\mathbf{X}^T - 2\sum_{l,p=1}^{L}(\widetilde{\mathbf{T}}^l_i)^T\widetilde{\mathbf{T}}^p_i\mathbf{X}\Theta\Theta^T,$$

which follows the second partial derivative of $f(\mathcal{E})$ with respect to $\mathrm{diag}(\mathcal{E})$,

$$\frac{\partial^2 f(\mathcal{E})}{\partial(\mathrm{diag}(\mathcal{E}))^2} = \sum_{l,p=1}^{L}(\widetilde{\mathbf{T}}^p_i)^T\widetilde{\mathbf{T}}^l_i\mathbf{X}\Theta\Theta^T\mathbf{X}^T. \tag{9}$$

As $\widetilde{\mathbf{T}}^p_i$ and $\widetilde{\mathbf{T}}^l_i$ are adjacency matrices of different layers in the model, which remain unchanged across different layers of the model, thus $\widetilde{\mathbf{T}}^p_i = \widetilde{\mathbf{T}}^l_i$. And we can rewrite (9) as

$$\frac{\partial^2 f(\mathcal{E})}{\partial(\mathrm{diag}(\mathcal{E}))^2} = \sum_{l=1}^{L}(\widetilde{\mathbf{T}}^l_i)^T\widetilde{\mathbf{T}}^l_i\mathbf{X}\Theta\Theta^T\mathbf{X}^T.$$

Since $A = (\widetilde{\mathbf{T}}^l_i)^T\widetilde{\mathbf{T}}^l_i$ and $(\mathbf{X}\Theta)(\Theta^T\mathbf{X}^T)$ are positive definite matrices, respectively, which follows that

$$(\widetilde{\mathbf{T}}^l_i)^T\widetilde{\mathbf{T}}^l_i\mathbf{X}\Theta\Theta^T\mathbf{X}^T$$

is positive definite. Therefore, $f(\mathcal{E})$ is a convex function, which can achieve the global optima.

V. Experiments

In this section, we evaluate and analyze the performance of our proposed method on six datasets commonly used for GNNs explanation. The evaluation methodology involves quantitative

TABLE I
THE STATISTICS OF ALL DATASETS

Datasets	Nodes	Edges	Labels
BA-House-Shapes	700	4110	4
BA-Community	1400	8920	8
BA-Grids	1020	5080	2
BA-Bottle-Shaped	700	3948	4
Tree-Cycle	871	1950	2
Tree-Grid	1231	3410	2

metrics and comparative analyses, providing a holistic view of the model's performance. We aim to offer insights into its robustness, reliability, and applicability across a spectrum of data types, ultimately contributing to a comprehensive understanding of its capabilities and limitations to subjecting the model to both artificial and real-world datasets.

A. Experimental Setup

Datasets: We employ six datasets to comprehensively explore the explanation of GNNs, four of which are real datasets and the other two are synthetic datasets. Real datasets include BA-House-Shapes, BA-Community, BA-Grids, and Tree-Cycles, and the synthetic datasets are Tree-Grids and BA-Bottle-Shaped, all of them are served as the pivotal benchmarks for assessing the explanation and performance of GNNs in various graph-based scenarios. Table I provides a detailed analysis of these datasets.

Baselines: We conduct a comprehensive comparison between our proposed model and nine commonly used GNN explanation methods on six datasets, that is, Dnx [26], GNNExplainer [9], PGExplainer [10], PGMExplainer [33], GraphMaskExplainer [11], GNN-LRP [38], GOAt [39], MixupExplainer [40] and GMT [37], which are described in detail in the Introduction. It strictly follows the same evaluation criteria in the same experimental environment to ensure the completeness and reliability of the comparisons. GNNExplainer is known for its interpretability capabilities in the GNN domain and is the benchmark against which our model performance is measured. PGExplainer provides instance-level explanations for GNNs by building a probabilistic graphical model. PGMExplainer is a typical method to explain GNN through surrogate models. In addition, GraphMaskExplainer explains GNN through predicting whether each edge in the graph can be discarded. GNN-LRP extends layer-wise relevance propagation to graphs, explaining predictions via higher-order relevant walks instead of individual nodes or edges. GOAt combines local and global attributions to uncover causal subgraphs, offering consistent multi-scale explanations for GNN predictions. MixupExplainer uses interpolation between graphs to estimate feature contributions, providing smoother and more robust instance-level explanations. GMT improves interpretability and generalization by approximating subgraph multilinear extensions through random subgraph sampling, achieving more faithful explanations than attention-based XGNNs. Although Dnx is also interpretable through a method based on knowledge distillation, it only considers the output of the last layer of the proxy model learning original model, and it ignores the knowledge learning for the intermediate feature map of the original model. The following are implementation details and evaluation metrics:

Implementation details: We conduct training sessions for a 3-layer GCN model, which is served as the original model for every individual dataset in our experiment. The training process employs the Adam optimizer with a learning rate set at 0.01. This approach ensures the effective training and optimization of the GCN model across all datasets, providing a solid foundation for subsequent analyses and evaluations.

In the knowledge distillation stage, we use a three-layer S ^{2} GC model as the proxy model, it involves leveraging the predictions of all nodes. We select a variety of commonly used loss functions as the distillation loss function of each layer to evaluate the our model's universality, applicability and robustness. Table II is obtained based on the *KL* divergence loss. We can effectively transfer the knowledge of the original model to the simpler proxy S ^{2} GC model through different loss functions, thereby achieving better results of knowledge distillation. As different loss functions may affect how the model learns, thereby impacting the quality of the explanations. We can understand which loss function is more conducive to improve the model's explanation performance under different distillation loss functions, it can help to select the most appropriate loss function for optimal explanation results in practical applications. This strategy improves the generalization performance of the model and ensures its accuracy and robustness.

Evaluation: We exploit multiple evaluation metrics to ensure a comprehensive and in-depth comparison. These metrics include accuracy of subgraph explanations, the overall runtime of the model, the accuracy of explained subgraphs under different inter-layer alignment-based loss functions and the Relative Improvement Rate (RIR), AUC, R-Fidelity and F-Fidelity. Below is the expansion on these evaluation metrics:

1) *Explanation Accuracy:* it characterizes the performance of original model on the explanation graph extracted by proxy model. Meanwhile, it is also a key metric to measure the mining ability of proxy model's on explanation graph. The effectiveness of the proxy model in revealing key factors can be accessed by comparing the consistency between the important subgraphs identified by the proxy model and the actual important ones. A higher accuracy of original model on explanation subgraph means the proxy model can more accurately identify the graph structures and node features that significantly impact the model's decisions.

2) *Distillation Time:* it is the time taken to distill the proxy model from the original model, which measures the distillation efficiency of the proposed model. The shorter the distillation time, the higher the efficiency of the distillation model.

3) *Overall Runtime:* it refers to all running time the model runs, including distillation time and explanation extraction time. Here, we calculate the running time of these methods to measure their efficiency, which is equally important in practical applications. A shorter overall runtime implies the model is more suited for those scenarios that require quick feedback,

TABLE II
COMPARISON OF EXPLANATION ACCURACY FOR DIFFERENT METHODS USING *KL* DIVERGENCE LOSS

Methods	BA-House-Shapes	BA-Community	BA-Grids	BA-Bottle-Shaped	Tree-cycle	Tree-grid	Average
DnX	0.6285 ± 0.06 7.75%	0.5031 ± 0.08 17.79%	0.7084 ± 0.07 4.91%	**0.5874** ± 0.03 -8.56%	0.7491 ± 0.05 -1.24%	0.6653 ± 0.02 1.02%	0.6403 3.13%
GNNExplainer	0.5237 ± 0.09 29.31%	0.4678 ± 0.05 26.68%	0.7194 ± 0.02 3.31%	0.4381 ± 0.06 22.60%	0.6865 ± 0.07 7.76%	0.5893 ± 0.03 14.05%	0.5708 15.69%
PGExplainer	0.5739 ± 0.03 18.00%	0.4538 ± 0.07 30.59%	0.7193 ± 0.08 3.32%	0.5268 ± 0.04 1.96%	**0.7580** ± 0.02 -2.40%	0.6352 ± 0.06 5.81%	0.6112 8.04%
PGMExplainer	0.5931 ± 0.08 14.18%	0.5785 ± 0.06 2.44%	0.4529 ± 0.09 64.10%	0.5714 ± 0.07 -6.00%	0.4827 ± 0.04 53.26%	0.4378 ± 0.03 53.52%	0.5194 27.13%
GraphMaskExplainer	0.5971 ± 0.05 13.41%	0.4716 ± 0.07 25.66%	0.7039 ± 0.09 5.58%	0.5042 ± 0.04 6.53%	0.7365 ± 0.06 0.45%	0.6654 ± 0.02 1.01%	0.6131 7.70%
GNN-LRP	0.5996 ± 0.05 12.94%	0.3377 ± 0.04 75.48%	**0.7462** ± 0.04 -0.40%	0.5226 ± 0.03 2.77%	0.7217 ± 0.08 2.51%	**0.7115** ± 0.06 -5.54%	0.6066 8.87%
GOAt	0.3095 ± 0.07 118.80%	0.1889 ± 0.02 213.64%	0.5950 ± 0.07 24.92%	0.2330 ± 0.02 130.50%	0.6310 ± 0.04 17.24%	0.6843 ± 0.06 -1.78%	0.4403 49.98%
MixupExplainer	0.5503 ± 0.04 23.06%	0.4239 ± 0.04 39.80%	0.5249 ± 0.05 41.57%	0.5335 ± 0.07 0.68%	0.6490 ± 0.03 13.99%	0.5643 ± 0.03 19.09%	0.5410 22.06%
GMT	0.5027 ± 0.21 34.71%	0.4836 ± 0.18 22.54%	0.5891 ± 0.14 26.16%	0.4909 ± 0.08 9.41%	0.7141 ± 0.09 3.60%	0.3471 ± 0.08 93.63%	0.5213 26.68%
Ours	**0.6772** ± 0.08	**0.5926** ± 0.06	0.7432 ± 0.03	0.5371 ± 0.05	0.7398 ± 0.07	0.6721 ± 0.04	**0.6603**

which is especially critical for large-scale graph data or real-time systems.

4) *Distillation Accuracy:* it mainly involves to the degree of prediction accuracy of that proxy models can achieve on the test set, while maintaining or improving the accuracy of the original model through relevant knowledge distillation. Distillation Accuracy can quantitatively analyze the learning level of the proxy model on the knowledge of the original model, ensuring the reliability of the proxy model in the knowledge distillation stage.

5) *RIR [28]:* is employed to measure the improvement of our model (A) over other models (B), the calculation is $RIR_B^A = \frac{Result(A)-Result(B)}{Result(B)} * 100\%$.

6) *Area Under Curve (AUC) [34]:* AUC measures how well an explainer ranks nodes that truly belong to the annotated key substructures. We treat node-importance scores as binary-classification scores and the dataset's important/non-important node labels as ground truth; sweeping a threshold yields the ROC curve, and its area is the AUC. Values range from 0 to 1; higher AUC indicates better prioritization of truly important nodes. Results can be aggregated as macro (per-graph AUC, then averaged) or micro (pool all nodes). Under severe class imbalance, PR-AUC may also be reported.

7) *Robust Fidelity (R-Fidelity) [36]:* R-Fidelity assesses explanation faithfulness under distribution shift by randomly thinning the explanation and non-explanation parts. It replaces the classic $\text{Fid}^{+}/\text{Fid}^{-}$ with α-controlled variants that sample edges from the two regions, yielding $\text{Fid}_{\alpha_1,+}$, $\text{Fid}_{\alpha_2,-}$, and $\text{Fid}_{\alpha_1,\alpha_2,\Delta} = \text{Fid}_{\alpha_1,+} - \text{Fid}_{\alpha_2,-}$. Tuning α limits OOD artifacts by controlling perturbation strength and keeps the metric well-behaved across scenarios.

8) *Fine-tuned Fidelity (F-Fidelity) [35]:* F-Fidelity strengthens fidelity evaluation in two coordinated steps: (i) explanation-agnostic fine-tuning with random masks so the model remains reliable on partially masked inputs, and (ii) matched stochastic masking at test time (capped by a budget β) to compute FFid^{+} and FFid^{-}. This pairing reduces information leakage and curbs OOD effects compared with removal-only metrics.

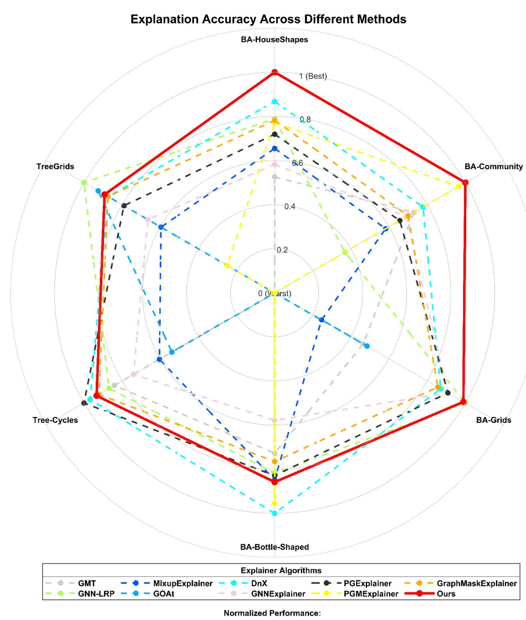


Fig. 3. Radar comparison maps of explanation accuracy across different methods on all data sets using *KL* divergence loss.

We can verify not only the theoretical performance of the model but also its potential and limitations in real-world application scenarios under considering these metrics together. This comprehensive evaluation approach helps us to gain a deeper understanding of the model's overall performance and guides future research directions and model optimization.

B. Explanation Performance

The explanation performance can be evaluated by Node Explanation Accuracy, its target is to explore the ability of the proxy model to find explanation node, and to characterize the error bounds of explanation node to original model.

1) Synthetic Dataset: Table II and Fig. 3 show a comprehensive comparison of Explanation Accuracy between our

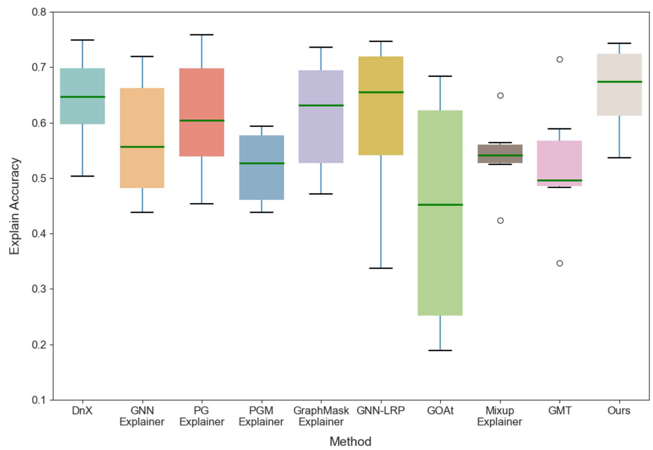
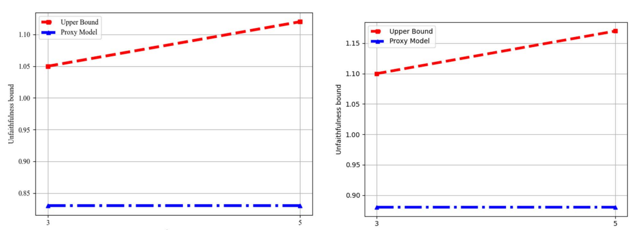


Fig. 4. Box-plot of explanation accuracy distribution across different methods using *KL* divergence loss.



(a) Unfaithfulness bound of the explanation with respect to original model. (b) Unfaithfulness bound of the explanation refers to proxy model.

Fig. 5. Unfaithfulness Bound of Explanations with Respect to Proxy and Original Models, Derived from Lemma 1 and Theorem 1.

method and nine other commonly used methods on six different database, which clear show that our method outperforms GNNExplainer on all six datasets. GNNExplainer is a popular method in the field, long recognized for its ability to generate high-quality explanations. However, our method surpasses it in terms of Explanation Accuracy, suggesting that our method can provide more precise and reliable explanations. Furthermore, our method consistently shows superior performance on all six datasets: it performs the best on the BA-House-Shapes and BA-Community, ranking second with a 0.003 gap on the BA-Grids datasets, ranking third on the BA-Bottle-Shaped, Tree-grid and Tree-cycle datasets. The average performance of Table II shows our method is the best one among DnX, GNNExplainer, PGExplainer, PGMExplainer, GraphMaskExplainer, GNN-LRP, GOAt, MixupExplainer and GMT, while it increases the subgraph Explanation Accuracy by 3.13%, 15.69%, 8.04%, 27.13%, 7.70%, 8.87%, 49.98%, 22.06% and 26.68%, respectively. Fig. 4 exhibiting the distribution of Explanation Accuracy of all algorithms, which clearly illustrates the central tendency and dispersion of the model's prediction. The results show our method can provide more accurate explanations.

In summery, the above results show the proxy model with a good ability in finding explanation subgraph. Fig. 5 gives the Explanation Accuracy of proxy and original models and their

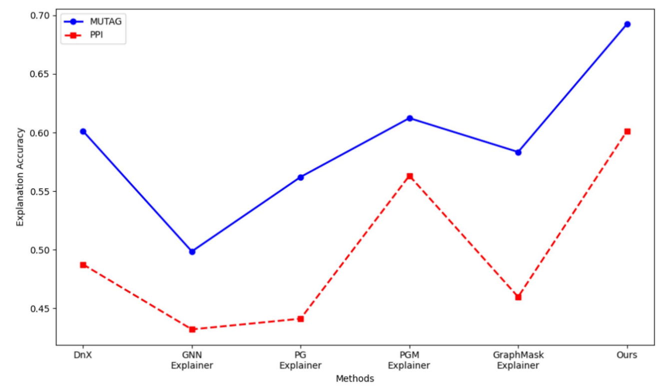


Fig. 6. Explanation accuracy for different distillation losses in real datasets.

upper bounds according to Lemma 1 and Theorem 1, respectively, which verifies the accuracy of the theoretical results, that is, the theoretical upper bounds we get in Lemma 1 and Theorem 1 are indeed the upper bounds of the empirical error of the proxy and original models.

It is worth mentioning that our method also shows significant advantages over Dnx on five of the six datasets. Although Dnx can provide highly accurate explanations, but it can't analyze what the proxy model can learn about the original model and how to learn from the original model, that is, it can not explore the distillation process of the model, only obtain the distillation results of the model. In summary, the analysis in Table II clearly demonstrates the superiority of our method in terms of Subgraph Explanation Accuracy compared with nine commonly used methods (i.e., Dnx, GNNExplainer, PGExplainer, PGMExplainer, GraphMaskExplainer, GNN-LRP, GOAt, MixupExplainer and GMT), which demonstrates its effectiveness and reliability in providing accurate and insightful explanations. It enables users to gain a deeper understanding of complex data and facilitate informed decision-making processes.

2) Real Dataset: Table IV shows a comprehensive comparison of Explanation Accuracy between our method and five other commonly used methods on two different real database. The table presents a comparison of explanation accuracy across various methods on two real-world datasets, MUTAG and PPI. Each method's performance is reported as a mean accuracy value accompanied by its standard deviation (e.g., 0.6012 ± 0.07 for DnX on MUTAG). Additionally, the percentage change relative to a baseline (likely a reference method) is provided in the second row for each method. For instance, DnX achieves a 15.19% improvement on MUTAG and a 23.32% improvement on PPI. Among the methods, Ours (likely the proposed method) demonstrates the highest accuracy on both datasets, with 0.6925 ± 0.04 on MUTAG and 0.6012 ± 0.09 on PPI. Other methods, such as GNNExplainer, PGExplainer, PGMExplainer, and GraphMaskExplainer, show varying levels of performance, with GNNExplainer exhibiting the largest percentage changes (38.89% on MUTAG and 39.13% on PPI). Overall, the average of Table IV highlights the comparative effectiveness of the

TABLE III
NODE EXPLANATION ACCURACY OF OUR METHOD UNDER DIFFERENT DISTILLATION LOSSES

	BA-House-Shapes	BA-Community	BA-Grids	BA-Bottle-Shaped	Tree-cycle	Tree-grid	Average
Cosine Similarity	0.6425	0.5760	0.7319	0.4570	0.7210	0.6870	0.6369
KL Divergence	0.6645	0.5815	0.7370	0.5190	0.7290	0.6880	0.6532
Consistency Loss	0.6455	0.5807	0.7313	0.4730	0.7250	0.6870	0.6404
Contrastive Loss	0.6475	0.5782	0.7310	0.4900	0.7217	0.6870	0.6426
MSE	0.6485	0.5765	0.7310	0.4765	0.7200	0.7010	0.6423

TABLE IV
COMPARISON OF EXPLANATION ACCURACY ON REAL DATASETS

Methods	MUTAG	PPI	Average
DnX	0.6012 $\pm$ 0.07 15.19%	0.4875 $\pm$ 0.09 23.32%	0.5444 18.83%
GNNExplainer	0.4986 $\pm$ 0.03 38.89%	0.4321 $\pm$ 0.05 39.13%	0.4654 39.00%
PGExplainer	0.5621 $\pm$ 0.03 23.20%	0.4412 $\pm$ 0.08 36.26%	0.5017 28.94%
PGMExplainer	0.6123 $\pm$ 0.08 13.10%	0.5632 $\pm$ 0.04 6.75%	0.5878 10.06%
GraphMaskExplainer	0.5834 $\pm$ 0.07 18.70%	0.4598 $\pm$ 0.05 30.75%	0.5216 24.01%
Ours	0.6925 $\pm$ 0.04	0.6012 $\pm$ 0.09	0.6469

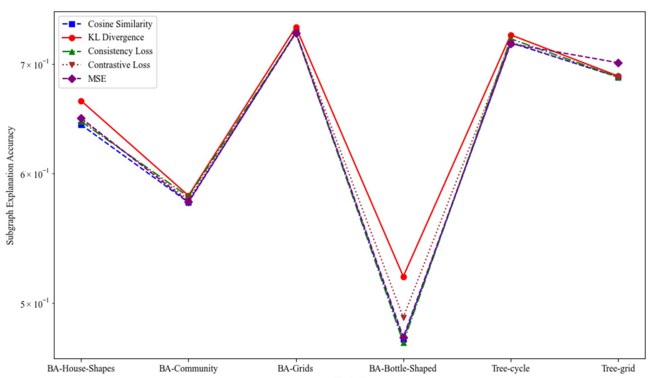


Fig. 7. Explanation Accuracy for Different Distillation Losses.

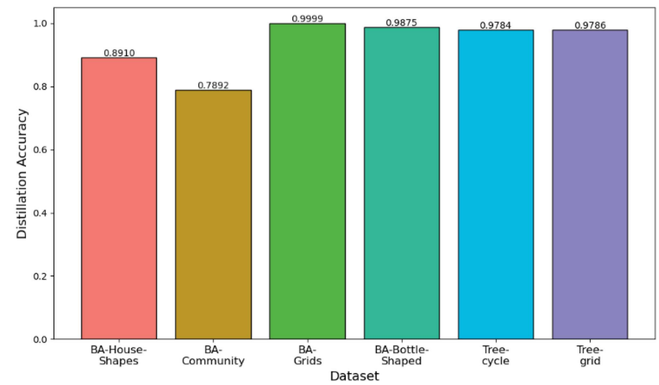


Fig. 8. Distillation Accuracy of Origin Model on Six Datasets.

TABLE V
EXPLANATION ACCURACY VERSUS MODEL'S DEPTH

	BA-Community	Tree-cycle	Tree-grid	Average
Dnx (3 layers)	0.5535	0.7220	0.6850	0.6535
Ours (3 layers)	0.5815	0.7290	0.6880	0.6662
Dnx (5 layers)	0.5343	0.6903	0.6621	0.6289
Ours (5 layers)	0.5890	0.7458	0.6902	0.6750

methods in terms of explanation accuracy and their relative improvements over the baseline.

C. Distillation Loss Selection

There are various methods to measure the distillation losses between the original model's latent layers and those of proxy model, here we incorporate five widely recognized distillation losses: Cosine Similarity, *KL* Divergence, Consistency Loss, Contrastive Loss, and Mean Square Error (MSE). The Subgraph Explanation Accuracy under the different distillation losses are shown in Table III and Fig. 7, from which we can see that the differences between the maximum and the minimum are no more than 0.006 (0.6485-0.6425), 0.0055, 0.006, 0.062, 0.009 and 0.014 on BA-House-Shapes, BA-Community, BA-Grids, BA-Bottle-Shapes, Tree-cycle and Tree-grid, respectively. The fluctuation of Table III is small, which demonstrate the influence of different distillation losses on the model' outputs is stable, and our model is universal, applicable and robust.

It is worth noting that the choice of distillation function plays a key role in determining the effectiveness of the entire distillation process. Cosine Similarity, *KL* Divergence, Consistency Loss, Contrastive Loss, and MSE all make unique contributions to the refinement of model explanation. A nuanced evaluation of these functions allows us to distinguish their individual effects on the explanation enhancement observed in knowledge distillation with inter-layer alignment. We can clearly see from Table III that no matter which kind of distillation losses is selected, the average performance is all improved compared to the nine commonly used methods for GNNs explanation that shown in Table II.

D. Model Layer Selection

The number of model layers is closely related to its explanatory power. Table V is obtained by employing a 3-layer GCN architecture for the original model and S^2GC network for the proxy model, and utilize the Kullback-Leibler (*KL*) divergence as the distillation loss function for inter-layer alignment-based distillation. The rationale behind this choice stems from our imperative requirements in solving the challenge of over-smoothing within the model architecture. Essentially, our goal is to elucidate how the inclusion of S^2GC can help alleviate problems associated with the smooth transition of information throughout the model. Our decision to integrate S^2GC into our methodology is based on the understanding that smooth transition is a key aspect of model explanation.

TABLE VI
THE DISTILLATION EFFICIENCY INCLUDE DISTILLATION ACCURACY AND DISTILLATION TIME OF OUR METHOD

Dataset	Distillation Accuracy	Distillation Times(s)
BA-House-Shapes	0.9339	15.6470
BA-Community	0.7866	8.9010
BA-Grids	0.9999	8.7960
BA-Bottle-Shaped	0.9875	10.2280
Tree-cycle	0.9827	9.7210
Tree-grid	0.9786	9.4030

TABLE VII
THE DISTILLATION ACCURACY FOR ORIGIN MODEL

Dataset	Distillation Accuracy
BA-House-Shapes	0.8910
BA-Community	0.7892
BA-Grids	0.9999
BA-Bottle-Shaped	0.9875
Tree-cycle	0.9784
Tree-grid	0.9786

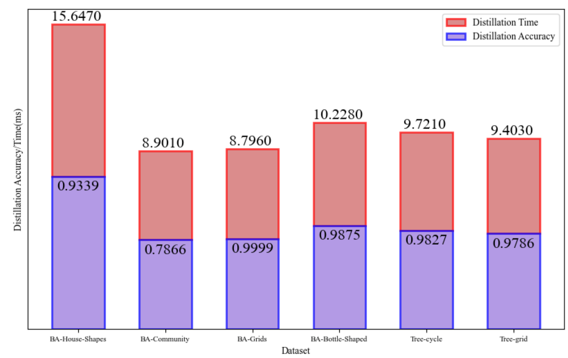


Fig. 9. Distillation Efficiency Include Distillation Time and Distillation Accuracy of Our Method on Six Datasets.

Traditional distillation methods, such as DnX, often suffer from a degradation in explanation performance as the model depth increases (see Table V). It is attributed to the increase in model complexity, making it increasingly difficult to maintain a coherent and smooth transition of information from one layer to the next. In contrast, our innovative approach exhibits unique incremental trajectories by considering S ^{2} GC as a proxy model, that is, subgraph explanation accuracy significantly improves as the model's depth increases (see Table V). This observation highlights a key advantage of our approach: it can effectively utilizes deeper architectural layer to enhance the model's interpretability. The relationship between model depth and explanation accuracy is a key aspect for our approach to successfully navigate.

The inherent challenges associated with deep models' explanation can be systematically addressed through the integration of S ^{2} GC with inter-layer alignment. It can not only solve oversmoothing, but also improve the model's explanatory ability of the model as depth increases, where inter-layer alignment is a crucial process in model optimization, aiming to transfer knowledge from a complex original model to a simpler proxy model. One of the main takeaways from our findings is the contrasting behavior of traditional approaches such as DnX versus our innovative approach: the explanation performance of DnX decreases with deeper architectures, and our approach exhibits a positive correlation between model depth and explanation accuracy. This dynamic highlights the effectiveness of our approach in leveraging the strengths of deeper models without compromising interpretability. Essentially, our approach leverages the depth of the model as an asset. This paradigm shift of leveraging deep architectures to improve interpretability places our approach at the forefront of advancements in distillation methods. The results presented in Table V serve as empirical evidence supporting the effectiveness of our chosen strategy, and reinforce the notion that our approach performs well in maintaining or even improving subgraph explanation accuracy as model depth increases.

E. Distillation Efficiency

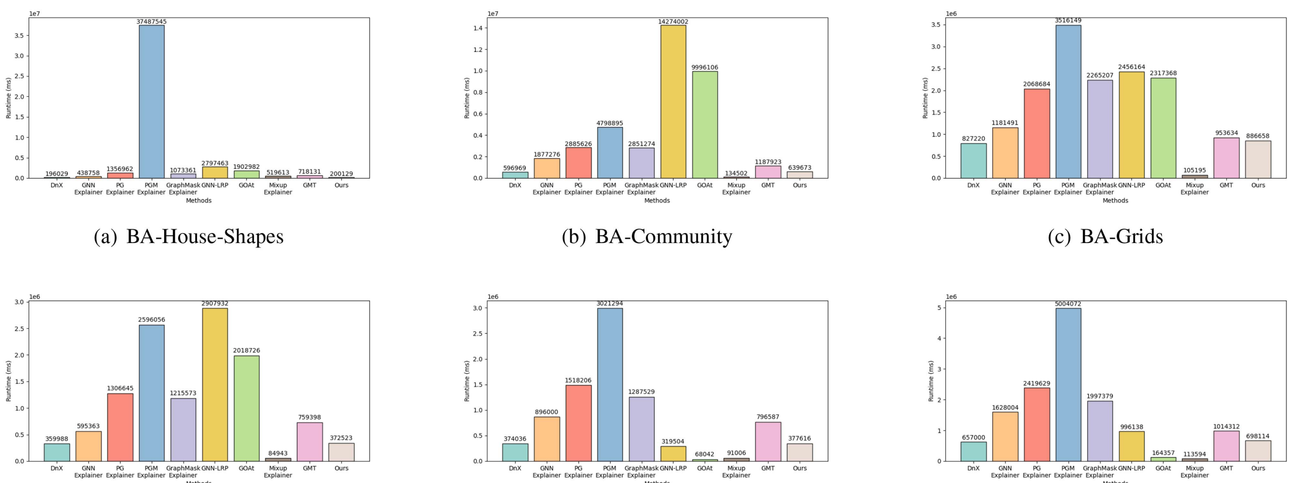
Distillation efficiency can be measured by Distillation Accuracy, Distillation Time and Overall Runtime, its primary purpose is to measure the accuracy and time of distilling the proxy model from the original model. We can obtain from Table VI that the proxy model achieves remarkable distillation accuracy rates, surpassing 93% on five out of six datasets and exceeding 97% on four datasets. Particularly noteworthy is its exceptional performance on the BA-Grids dataset, where the distillation accuracy approaches the theoretical limit of 1. Despite a slightly lower accuracy on the BA-Community dataset, registering at 78%, our method demonstrates its ability to impart substantial knowledge from the original model to ensure the proxy model's reliability.

The distillation time in Table VI reveals that the longest duration is approximately 15.647 seconds, while most of the times are within the 10-second range. Impressively, the shortest time recorded is merely 8.796 seconds, we complete the entire distillation process with a remarkable 99% accuracy for the proxy model. The results in Table VI and Fig. 9 show the time spends on model distillation is negligible, which reinforces the efficiency of the distillation method with inter-layer alignment and greatly reduces the burden of the distillation model on the entire model runtime, allowing the model to spend more time on explanation extraction.

Table VIII and Fig. 10 provide a comprehensive comparison with respect to the overall explanation time of our model with other four explainable methods: GNNExplainer, PGExplainer, PGMExplainer and GraphMaskExplainer on six datasets. Table VIII demonstrates the average Overall Runtime of our method is much smaller than those of the other three methods except for DnX and MixupExplainer, it decreases the running time by 53.53%, 73.58%, 94.67%, 71.58%, 87.29%, 81.61% and 42.95%, respectively, while it slightly increases by 5.92% and 20.45% compared with DnX and MixupExplainer, respectively. However, the average explanation performance of our proposed method increases by 3.13% and 22.06% in the Table II compared with DnX and MixupExplainer, respectively. Therefore, it is reasonable and acceptable to tolerate a slightly higher complexity under a certain error range in the era of high performance computing.

TABLE VIII
COMPARISON OF OVERALL RUNTIME ACROSS ALL METHODS ON SIX DATASETS, INCLUDING DISTILLATION TIME AND FINAL MODEL EVALUATION TIME(MS)

	BA-House-Shapes	BA-Community	BA-Grids	BA-Bottle-Shaped	Tree-cycle	Tree-grid	Average
DnX	166029	566969	797220	329988	344036	627000	471874
	-2.47%	-7.54%	-7.45%	-3.80%	-1.04%	-6.56%	-5.92%
GNNExplainer	408758	1847276	1151491	565363	866000	1598004	1074815
	58.38%	67.01%	25.59%	39.42%	59.86%	58.18%	53.53%
PGExplainer	1326962	2855626	2038684	1276645	1488206	2389629	1891959
	87.17%	78.65%	57.98%	73.17%	76.62%	72.04%	73.58%
PGMExplainer	37457545	4768895	3486149	2566056	2991294	4974072	9373185
	99.55%	87.21%	75.43%	86.64%	88.38%	86.57%	94.67%
GraphMaskExplainer	1043361	2821274	2235207	1185573	1257529	1967379	1758387
	83.69%	78.40%	61.67%	71.10%	72.35%	66.03%	71.58%
GNN-LRP	2767463	14244002	2426164	2877932	289504	966138	3928533
	93.85%	95.71%	64.69%	88.09%	-20.07%	30.84%	87.29%
GOAt	1872982	9966106	2287368	1988726	38042	134357	2714596
	90.91%	93.88%	62.54%	82.77%	-813.76%	-397.268%	81.61%
MixupExplainer	489613	1045020	751950	54943	61006	83594	414354
	65.25%	41.65%	-13.92%	-523.41%	-469.80%	-699.23%	-20.45%
GMT	688131	1157923	923634	729398	766587	984312	874997
	75.27%	47.34%	7.25%	53.04%	54.65%	32.12%	42.95%
Ours	170129	609673	856658	342523	347616	668114	499789



(a) BA-House-Shapes (b) BA-Community (c) BA-Grids

(d) BA-Bottle-Shaped (e) Tree-cycle (f) Tree-grid

Fig. 10. Comparison of Overall Runtime for All Methods Across Six Datasets, Including Total Execution Time for Training and Inference.

The results show the significant efficiency of our method, it significantly faster than GNNExplainer, PGExplainer, PGMExplainer, GraphMaskExplainer, GNN-LRP, GOAt and GMT on all datasets. This advantage stems from the simpler proxy model we adopt, its high efficiency in knowledge distillation and the explanation extraction leads to a reduction in overall explanation time. Additionally, our method consumes slightly more time than DnX, because DnX only focuses on the alignment of the last layer of the original model, while we choose to learn the intermediate output for each layer.

F. Interpretability Measurement

Table IX presents the comparison results between the DnX and our method across different synthetic datasets, and the evaluation metrics include R-Fidelity, F-Fidelity, and AUC. The results indicate that our proposed method outperforms DnX on most datasets across all three metrics, further demonstrating superior interpretability and robustness. Particularly, the outstanding results on R-Fidelity and F-Fidelity clearly indicate that the proposed inter-layer alignment distillation for subgraph extraction indeed achieves higher credibility compared to conventional ones, thereby further validating the effectiveness of the proposed distillation approach in acquiring reliable subgraphs.

The values of R-Fidelity and F-Fidelity in the Table IX refer to the interpretability analysis of graph neural networks (GNNs), showing the performance changes of the model when specific features are retained and removed. Through these statistics, we can see the importance of the features extracted by the interpretation method to the model prediction, and the degree of model dependence on these features. These results provide a basis for evaluating the effectiveness of the interpretation method, and also reflect the potential optimization space of the

TABLE IX
THE PERFORMANCE COMPARISON OF DNX AND OUR METHOD IN TERMS OF R-FIDELITY, F-FIDELITY, AND AUC

Dataset	DnX			Ours		
	R-Fidelity↑	F-Fidelity↑	AUC ↑	R-Fidelity↑	F-Fidelity↑	AUC↑
BA-HouseShapes	0.1092±4.26E-05 3.21%	0.4893±0.07 18.62%	0.8365±0.0019 7.27%	**0.1127**±7.33E-06	0.5804±0.02	0.8973±0.0065
BA-Community	0.0470±9.24E-06 8.51%	0.2209±0.08 53.96%	0.7810±0.0010 9.50%	0.0510±6.07E-07	0.3401±0.06	0.8552±0.0009
BA-Grids	0.0247±3.54E-06 2.02%	0.4801±0.03 11.23%	0.9034±0.0005 1.43%	0.0252±4.99E-05	0.5340±0.05	**0.9163**±0.0003
BA-Bottle-Shaped	0.0222±4.47E-07 -3.45%	0.5200±0.03 36.25%	0.8110±0.0016 7.19%	0.1075±1.38E-06	**0.7085**±0.08	0.8693±0.0007
Tree-Cycles	-0.1143±3.10E-06 12.25 %	0.1203±0.04 110.39%	0.7093±0.0018 2.13%	0.0214±8.02E-05	0.2531±0.03	0.7244±0.0069
TreeGrids	0.0952±2.81E-05 12.99%	0.4838±0.07 -1.53%	0.7171±0.0029 2.69%	-0.1283±8.17E-06	0.4764±0.04	0.7364±0.0014
Average	0.0306±4.47E-07 3.04 %	0.3857±0.05 24.97%	0.7931±0.0028 5.06%	**0.0316**±2.46E-05	**0.4820**±0.05	**0.8332**±0.0016

TABLE X
THE EXPLANATION ACCURACY RANKINGS OF ALL ALGORITHMS

DnX	GNNExplainer	PGExplainer	PGMExplainer	GraphMaskExplainer	GNN-LRP	GOAt	MixupExplainer	GMT	Ours
3.00	6.6700	4.83	6.33	5.00	4.17	8.00	7.33	7.50	2.17

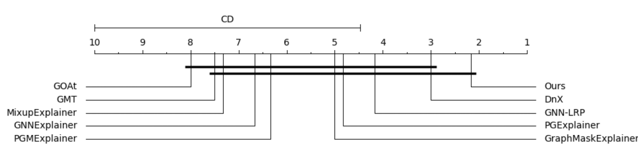


Fig. 11. The Nemenyi Test for All Algorithms.

model in terms of interpretability. Overall, the table provides strong support for understanding the model prediction behavior and its feature dependencies.

G. Statistical Test

This subsection presents the results of the Friedman test and post-hoc tests that are performed to evaluate and compare the performance of all algorithms. The Friedman statistic is expressed as $\chi_F^2 = \frac{12\,N}{k(k+1)}(\sum_{j=1}^{k} R_j^2 - \frac{k}{4}(k+1)^2), F_F = \frac{(N-1)\chi_F^2}{N(k-1)-\chi_F^2}$, where N refers to the total number of datasets, k denotes the number of algorithms compared, R_j represents the mean ranking of each algorithm across all datasets. F_F reproduces the test statistic for the Friedman test. Additionally, the critical difference is represented as $CD_\alpha = q_\alpha\sqrt{\frac{k(k+1)}{6N}}$, which is determined using the Nemenyi post-hoc test with a significance level of α, and α is typically set to 0.05. The critical value is represented as q_α, which depends on the sum of the algorithms and datasets.

The average ranking results refer to Explanation Accuracy of nine algorithms across six datasets can be seen Table X, it follows that the value of x_F^2 is 23.2167 and F_F is 3.7709. When $\alpha = 0.05$, the critical value of $x_F^2(10) = 18.307$ and $F(9, 45) = 2.096$, which are smaller than those of $x_F^2 = 23.2167$ and $F_F = 3.7709$, respectively. Therefore, there are significant differences between our method and other ones. Furthermore, charts are also used to further evaluate the differences between these algorithms. The comparison between our algorithm and other algorithms is shown in Fig. 11.

VI. CONCLUSION

We propose a holistic framework for the explanation of complex GNN model by using a simpler proxy model. In this framework, the complex nonlinear GNNs model can be explained or replaced by a simple linear model. Meanwhile, the ability of the proxy model to extract explanation important nodes can be measured quantitatively and theoretically, thereby ensuring the explanation of the original model. The advanced nature of our proposed method can be reflected from two aspects: the knowledge distillation and the explanation extraction. In the knowledge distillation stage, we employ an advanced inter-layer alignment technique to guarantee the accuracy and completeness of the distilled knowledge by considering both the topological structure and knowledge representation. In the explanation extraction, we theoretically characterize the faithfulness bound of the proxy model's reliability in extracting explanation subgraphs. In terms of experimental validation, we also design comprehensive experiments to verify the above ideas and the proposed method, which mainly include the effectiveness of extract explanation subgraphs with the proxy model, the faithfulness and efficiency of the knowledge distillation between original model and proxy model. The results of both quantitative experiments and qualitative theory demonstrate that the knowledge distillation with inter-layer alignment has significant advantages over the existing GNN explainable methods in sloving over-smoothing. Therefore, the proposed framework is flexible and broadly suitable for GNNs interpretability.

Although our proposed approach has made significant strides, enhancing model interpretability and demonstrating superior performance, it has not fully considered the unique topological structure of graph data, which encompasses complex relationships between nodes and the global information of the graph. Topological structure of graph data is crucial for a complete understanding and explanation of the decision-making processes of GNN models. Therefore, our future work will focus on integrating and utilizing the topological information to explore what and how the proxy model has learned from the original model. We work on the analytical distillation process, that is, the learning process of the proxy model from the original model, providing more reliable and transparent decision-support tools to practical applications in related fields.

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